

# Miles Jennings

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## EDUCATION

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### University of California Irvine

Irvine, CA

*Bachelor of Science in Computer Engineering*

*Sept. 2022 – June 2026*

**Relevant Coursework:** Organization of Digital Computers, Electronics I–II, Network Analysis I–II, Data Structures and Algorithms, Discrete and Continuous-Time Signals, Digital Logic, Computer Architecture, VLSI, Embedded Systems, Digital Signal Processing, Analog IC Design

**Organizations:** HyperXite, FUSION, FASAE

## EXPERIENCE

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### P2S Technology Intern

Long Beach, California

*P2S Engineering*

*June 2025 – Present*

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Supported engineers with **CAD** layout updates, wiring diagrams, and equipment schedules using AutoCAD and Revit
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs

### HyperXite – SpaceX Hyperloop

Irvine, California

*Control Systems Engineer*

*Aug. 2025 – Present*

- Designed, tested, and optimized the Hyperloop pod's control system hardware using an STM32H753ZI microcontroller programmed in **C/C++**, interfacing with thermistors, time-of-flight sensors, and INA219 current sensors while evaluating components for performance and cost improvements over prior pod iterations
- Ensured hardware–software compatibility by defining system constraints and key telemetry parameters based on feedback from team surveys identifying the most critical data points to track in the GUI
- Implemented a **Continuous Integration (CI) pipeline** using GitHub Actions to streamline testing, version control, and code deployment across subteam

## PROJECTS

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### MIPS & Pipeline Processor | *Verilog*

Jan. 2025 – Feb. 2025

- Designed and simulated a MIPS and pipelined processor using Verilog, ensuring correct instruction execution
- Implemented multiple MUX's, ALU-Controllers, and other combinational circuit components
- Developed and tested ALU controllers, multiplexers (MUXs), and other combinational logic components, optimizing data flow efficiency

### Dictionary & Calculator Program | *Java, Swing*

Sept. 2024 – Dec. 2024

- Developed a dictionary application supporting import/export of text files for word and definition management
- Developed a feature-rich calculator with event-driven programming, handling factorial, logarithmic, and memory functions

### Chess & Poker Program | *C, Linux, Git, SDL, GTK*

April 2024 – June 2024

- Developed a multiplayer Chess and Poker game in **C**, implementing server-to-client socket communication for real-time gameplay
- Utilized **Linux** for version control and coordination among team members.
- Designed an interactive GUI using SDL 1.2 (Chess) and GTK (Poker), enhancing user experience and engagement

### AudioVisor | *Java, PDM, nRF52840*

Sept. 2025 – Present

- Built an audio-reactive visor using the Adafruit nRF52840, using **Java** to receive and handle microphone input through **PDM** (Pulse Density Modulation) communication for real-time sound magnitude detection
- Prototyped and iterated hardware designs, including LED driver layouts and signal pathways, while troubleshooting performance issues throughout development
- Researched and evaluated components to satisfy sensitivity, latency, and power needs, guiding system architecture and part selection

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, Java, Verilog, RISC-V, MIPS Assembly, MATLAB

**Software/Tools:** Revit, AutoCAD, Bluebeam, Arduino IDE, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm